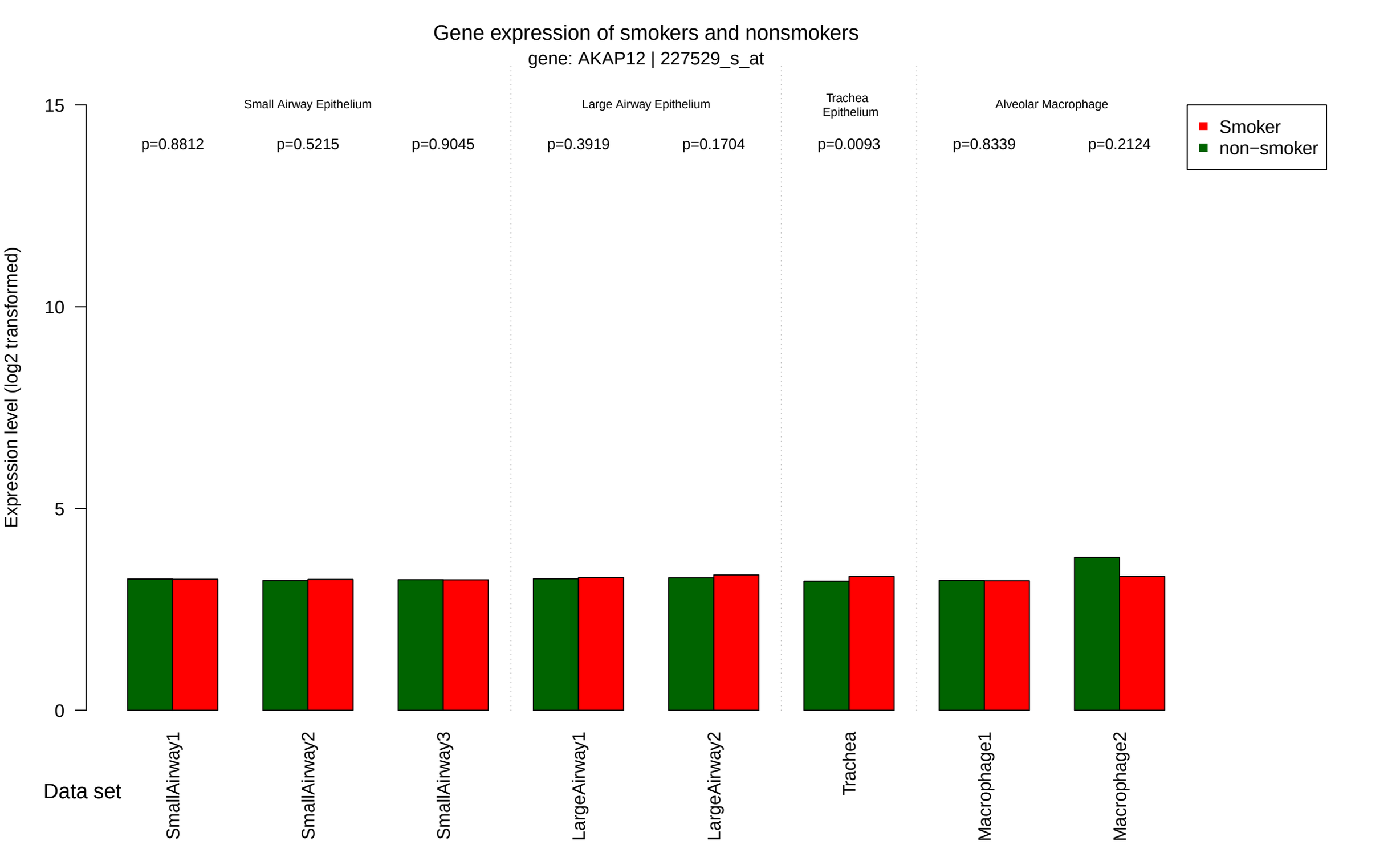
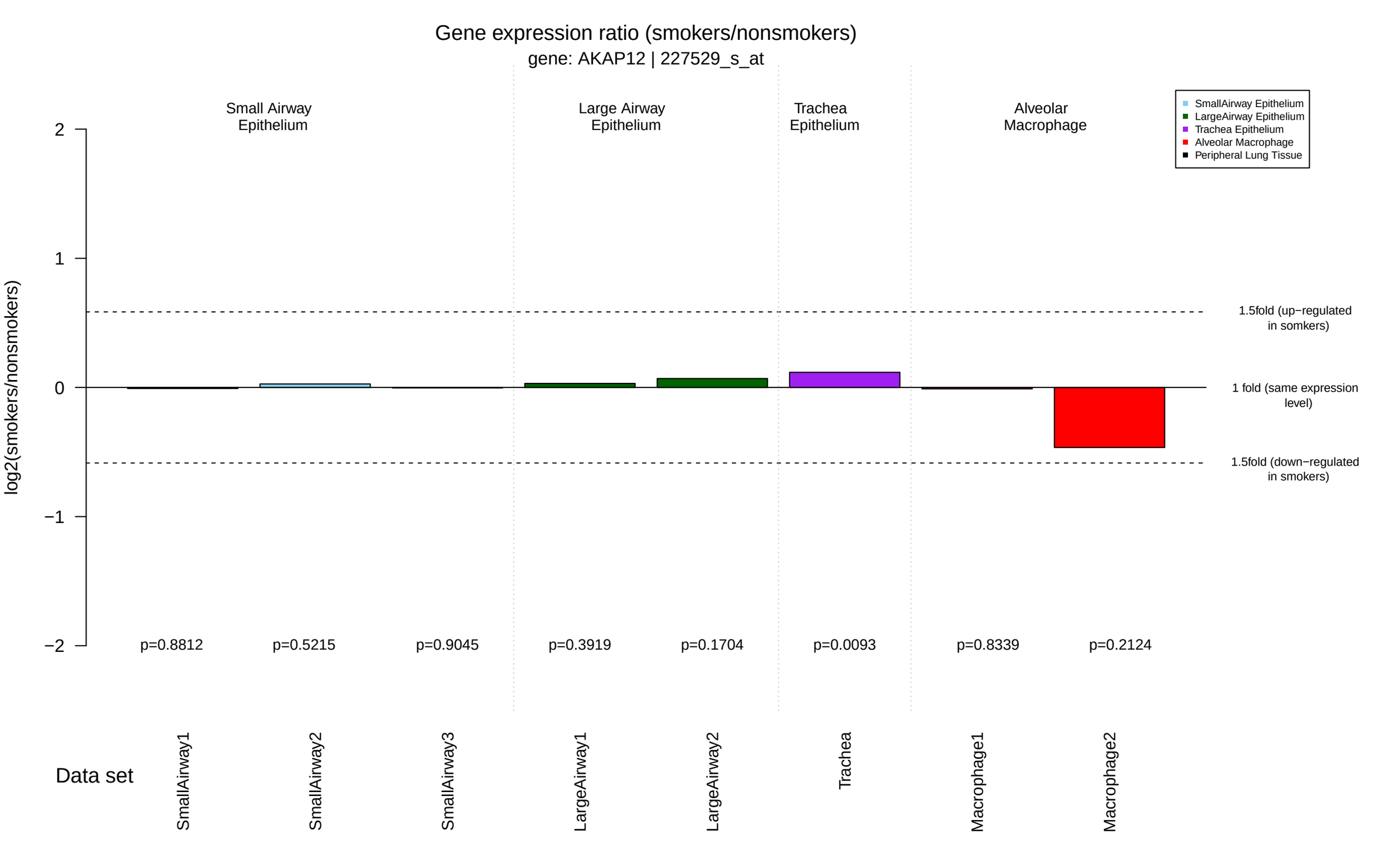
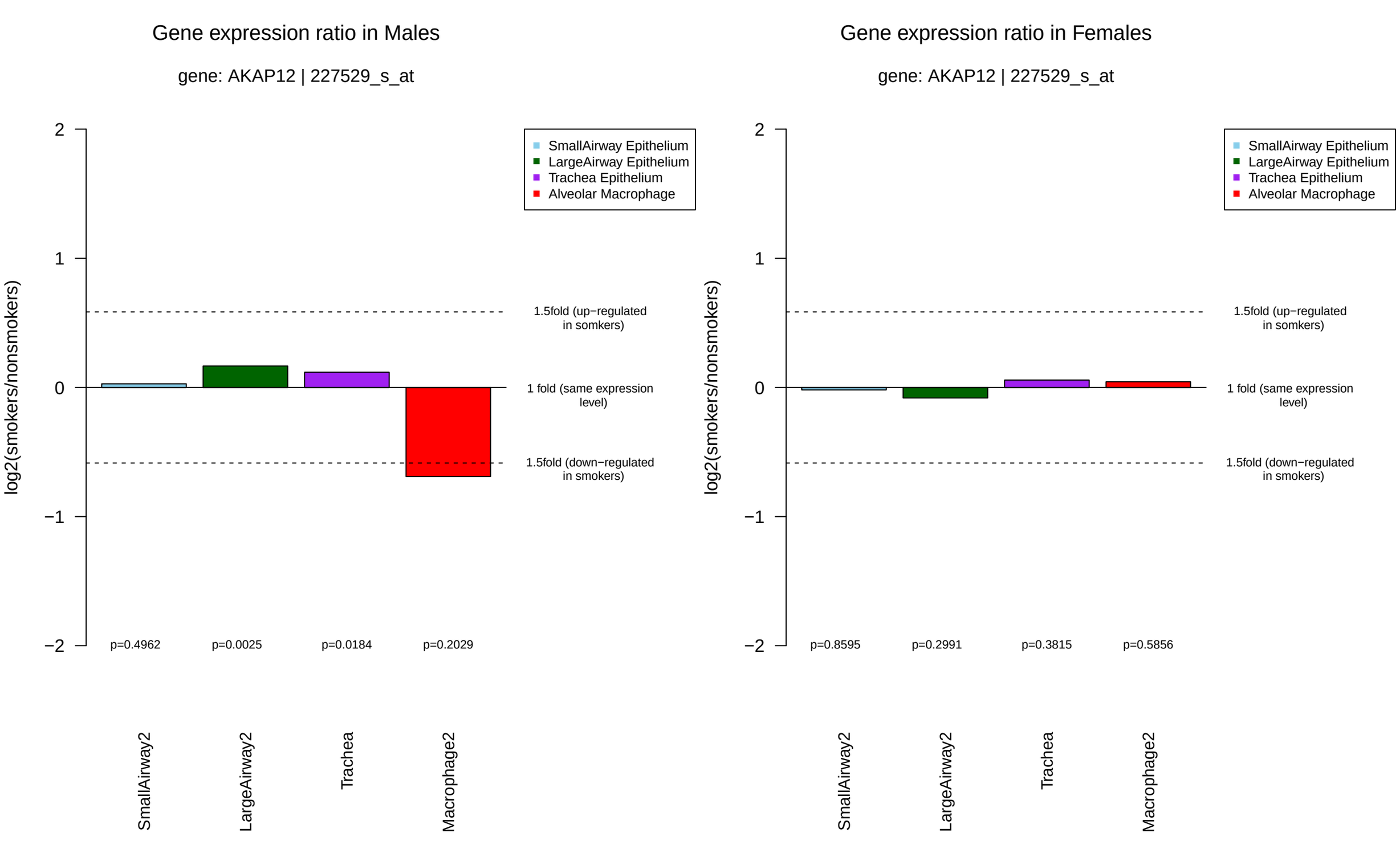
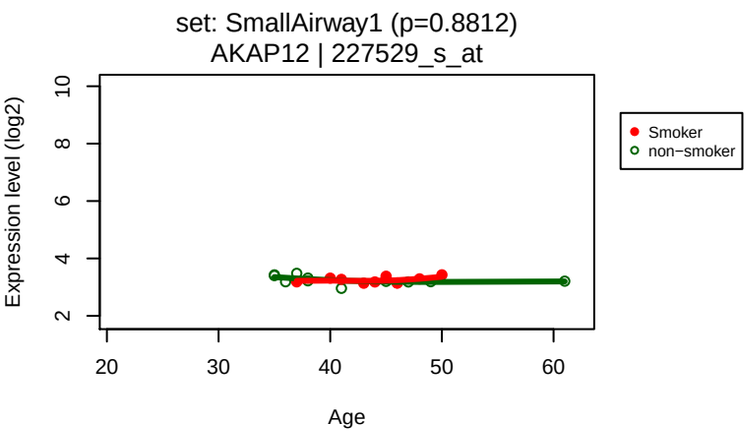
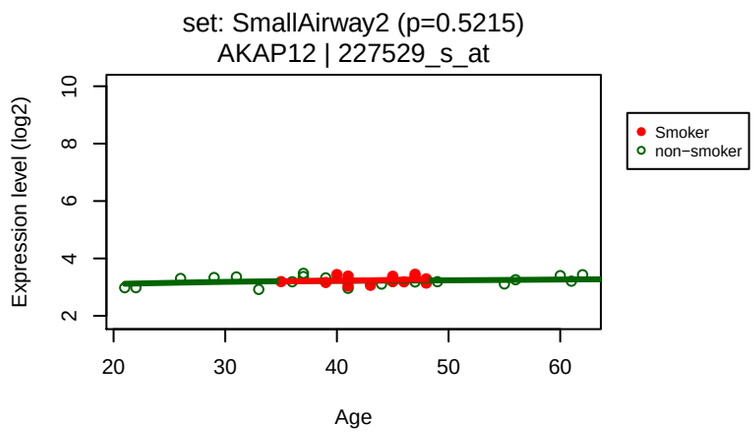
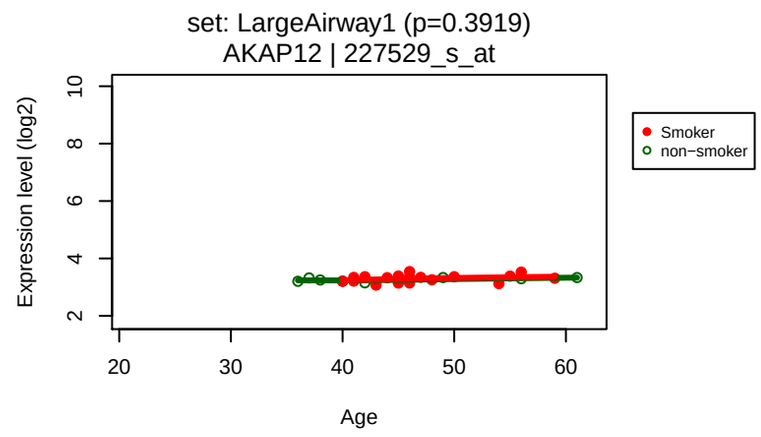

Gene expression of smokers and nonsmokers
gene: AKAP12 | 227529_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8812
p=0.5215
p=0.9045
p=0.3919
p=0.1704
p=0.0093
p=0.8339
p=0.2124
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

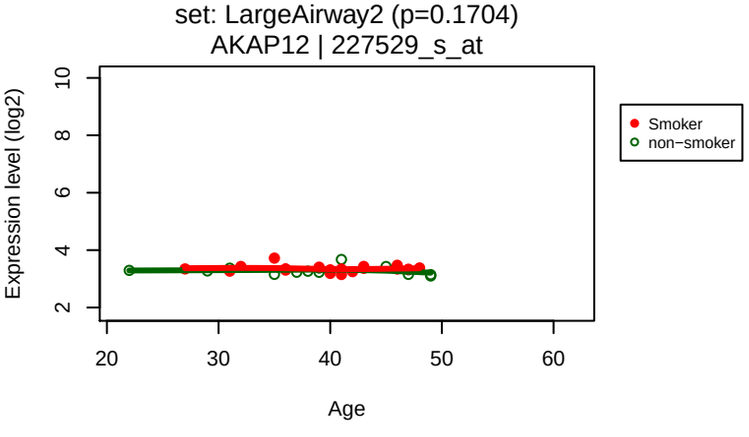

Gene expression ratio (smokers/nonsmokers)
gene: AKAP12 | 227529_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8812
p=0.5215
p=0.9045
p=0.3919
p=0.1704
p=0.0093
p=0.8339
p=0.2124
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

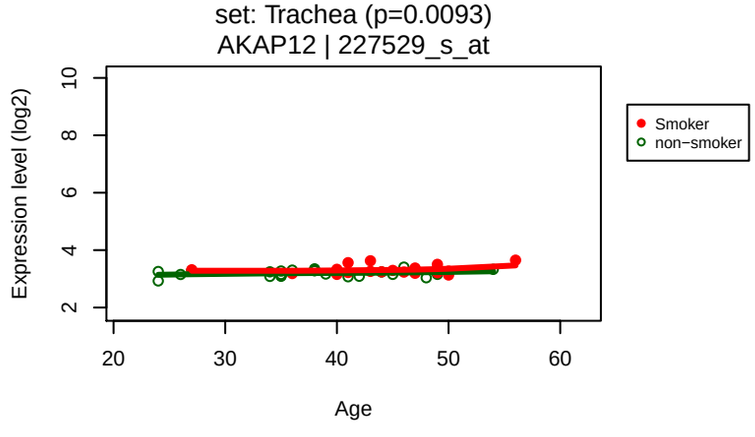

Gene expression ratio in Males
gene: AKAP12 | 227529_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4962
p=0.0025
p=0.0184
p=0.2029
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AKAP12 | 227529_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8595
p=0.2991
p=0.3815
p=0.5856
SmallAirway2
LargeAirway2
Trachea
Macrophage2

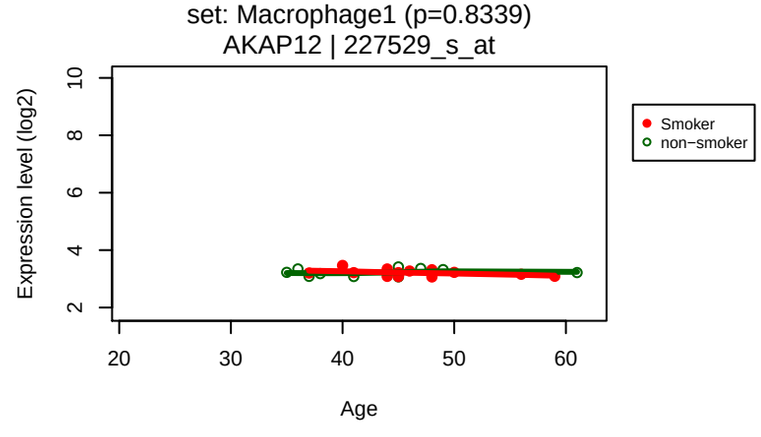

set: SmallAirway1 (p=0.8812)
AKAP12 | 227529_s_at
Expression level (log2)
Age
Smoker
non−smoker

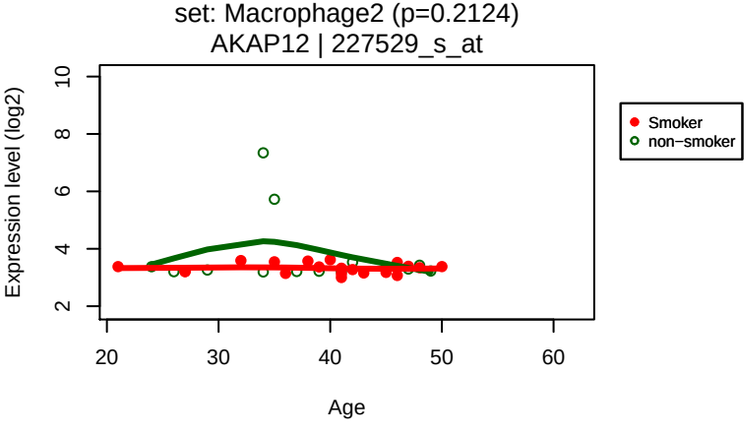

set: SmallAirway2 (p=0.5215)
AKAP12 | 227529_s_at
Expression level (log2)
Age
Smoker
non−smoker

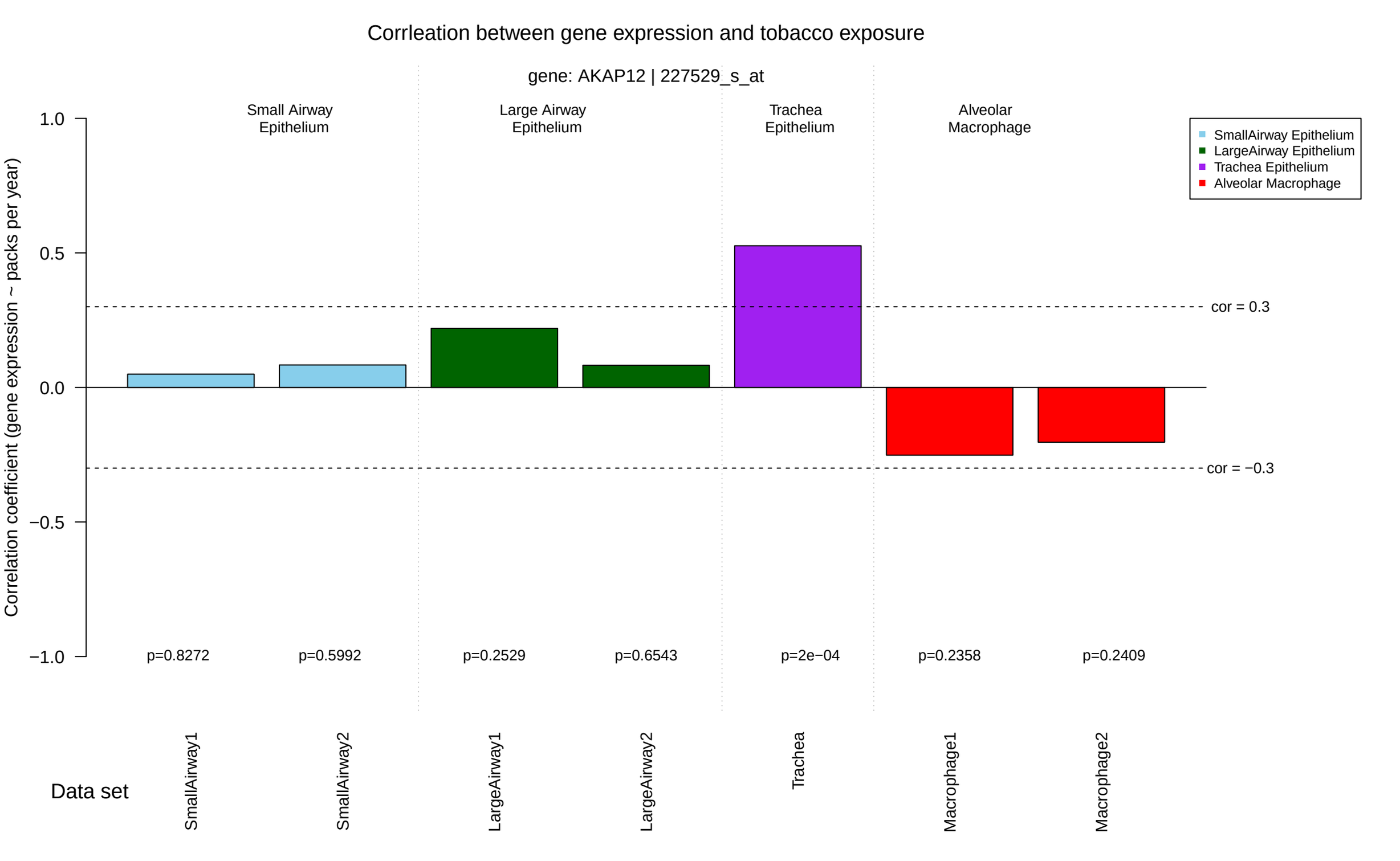

set: LargeAirway1 (p=0.3919)
AKAP12 | 227529_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1704)
AKAP12 | 227529_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0093)
AKAP12 | 227529_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8339)
AKAP12 | 227529_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2124)
AKAP12 | 227529_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AKAP12 | 227529_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8272
p=0.5992
p=0.2529
p=0.6543
p=2e−04
p=0.2358
p=0.2409
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2